Lecture:

PROTEIN EXPRESSION SYSTEMS

BIO 208

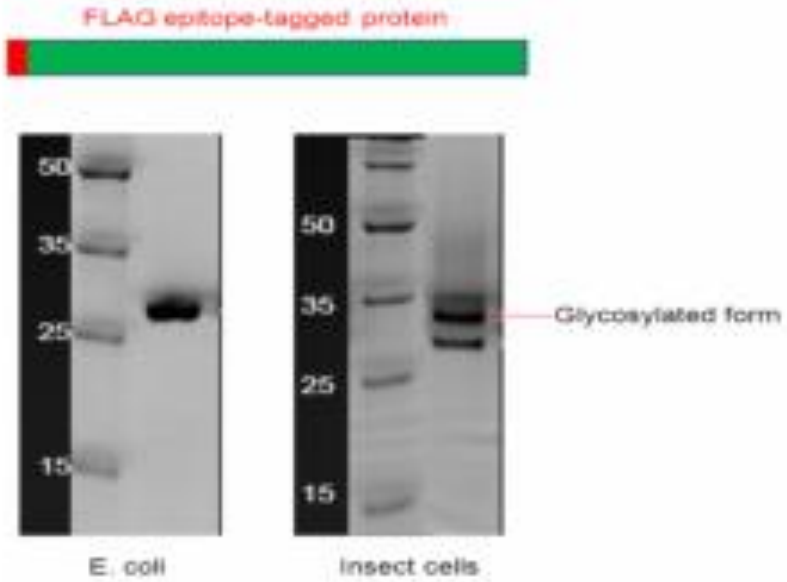
Bacterial expression systems

Advantages

- Grow quickly (8-12 hrs to produce protein)
- High yields (50-500 mg/L)
- Low cost of media
- Low fermentor costs

Disadvantages

- Difficulty expressing large proteins (>50 kD)
- Eukaryotic proteins are sometimes toxic
- Can't handle disulphide-bonded proteins
- No glycosylation or signal peptide removal



Baculovirus Systems

Advantages

- Can express large proteins (>50 kD)
- Correct glycosylation & signal peptide removal
- Has chaperonins to help fold “tough” proteins
- Very high yields, cheap

Disadvantages

- Grow very slowly (10-12 days for set-up)
- Cell culture is only sustainable for 4-5 days
- Set-up is time consuming, not as simple as yeast

Mammalian Systems

Advantages

- Can express large proteins (>50 kD)
- Correct glycosylation & signal peptide removal, generates authentic proteins
- Has chaperonins to help fold "tough" proteins

Disadvantages

- Selection takes time (weeks for set-up)
- Cell culture is only sustainable for limited period of time
- Set-up is very time consuming, costly, modest yields

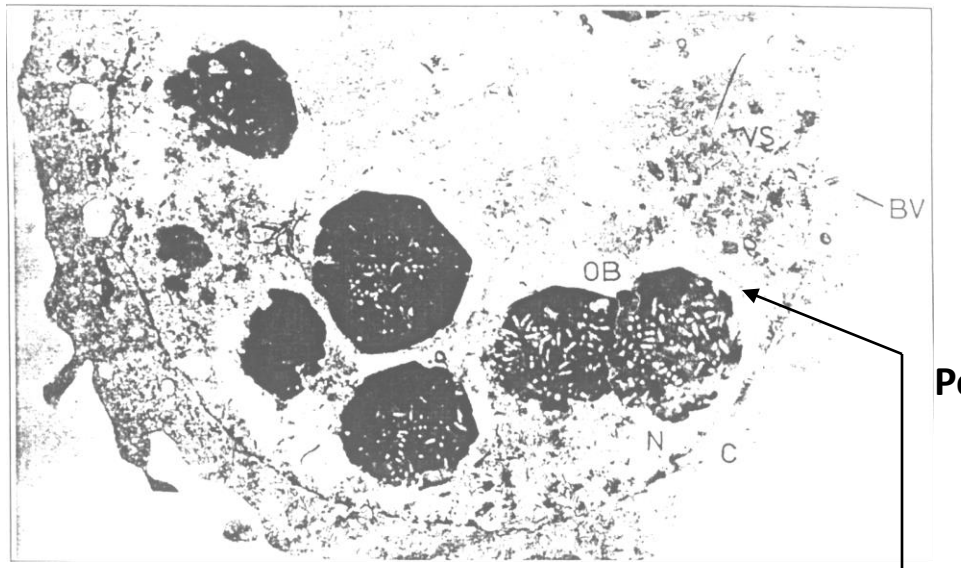
Yeast expression systems

Advantages

- Grow quickly (12-24 hrs to produce protein)
- Very high yields (50-5000 mg/L)
- Low cost of media (simple media constituents)
- Low fermentor costs
- Can express large proteins (>50 kD)
- Glycosylation & signal peptide removal
- Has chaperonins to help fold “tough” prtns
- Can handle S-S rich proteins

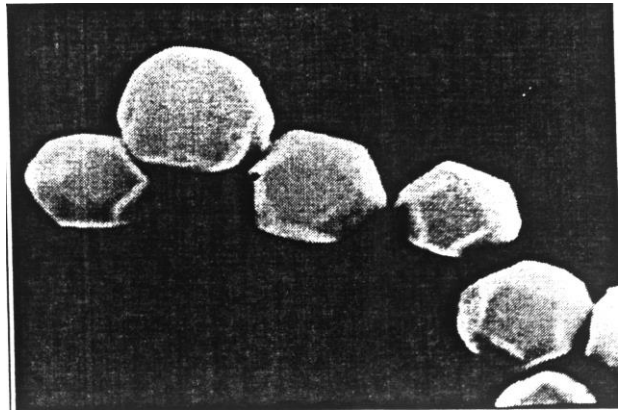
BACULOVIRUS EXPRESSION VECTOR SYSTEM

……..A tool to study proteomics

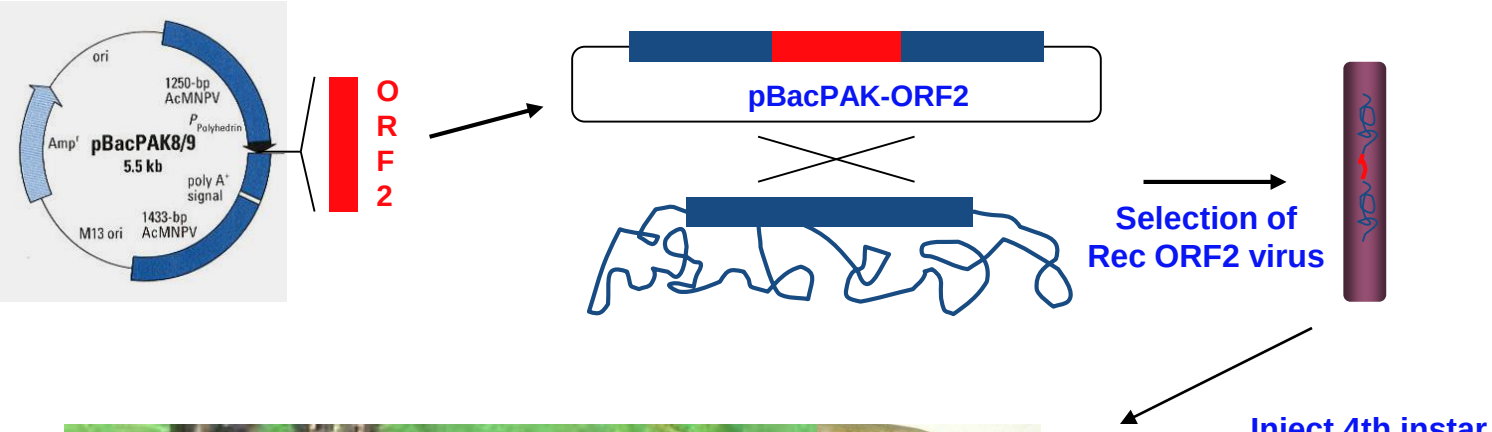


Polyhedra

Viral Capsids

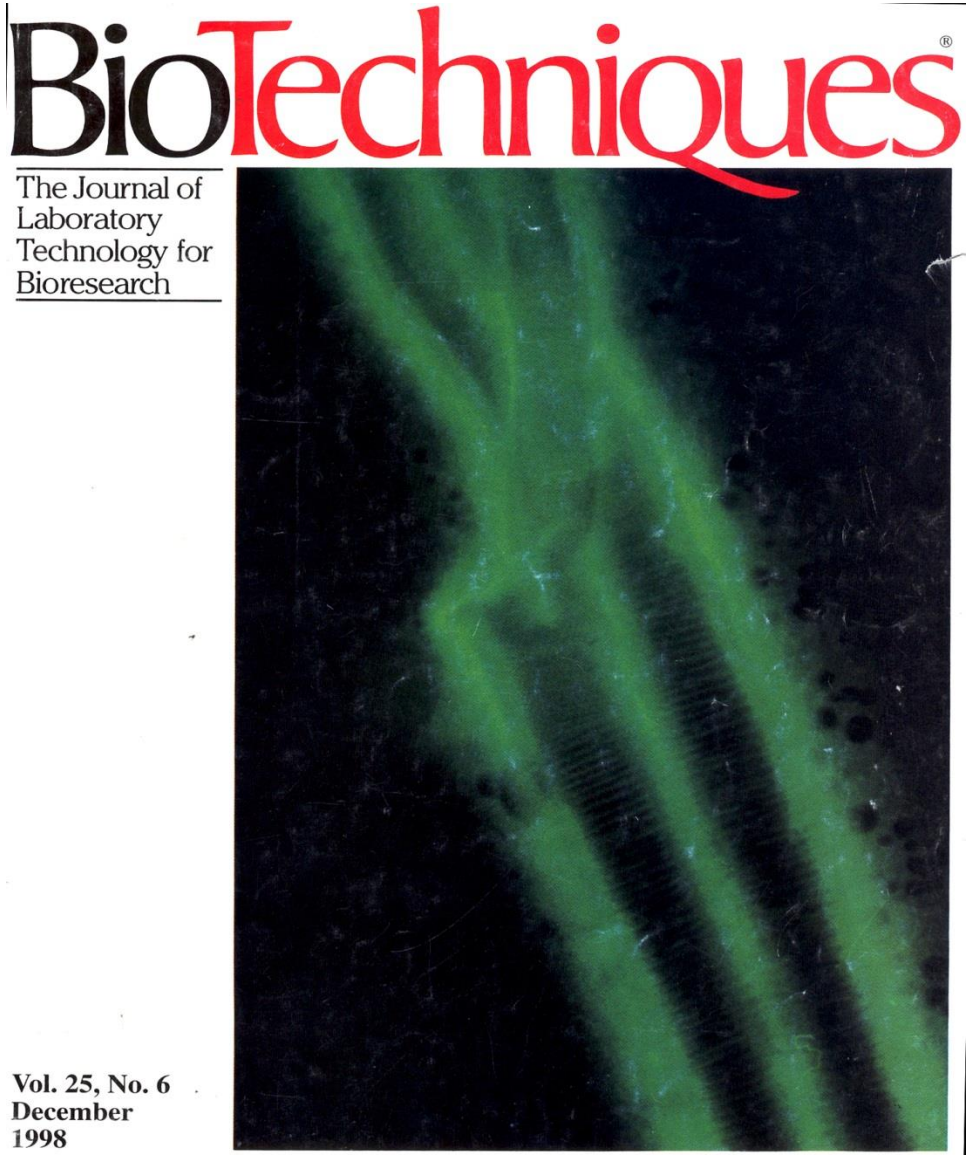


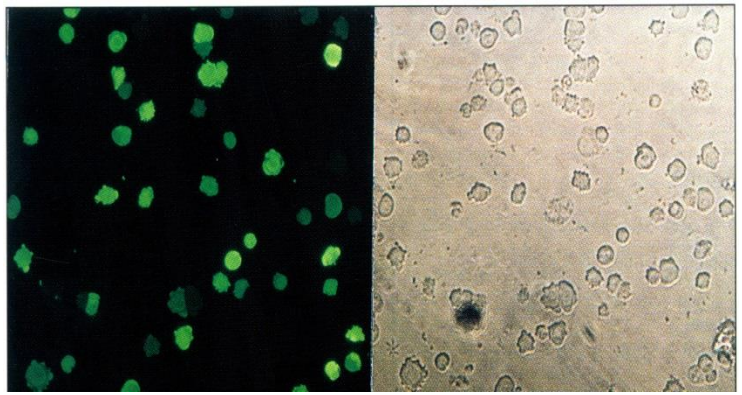
Generating recombinant virus and larval infection



Inject 4th instar S. litura larvae

Insects as Bioreactors for Protein drugs synthesis





WHY PURIFY PROTEINS

Protein purification is a series of processes intended to isolate a single type of **Protein** from a complex mixture.

Why purify proteins?

- Detailed studies on function
- Determination of structure
- Industrial/pharmaceutical applications
- Generate antibodies
- Amino acid sequence determination

16) Proteins separation can be carried out on the basis of

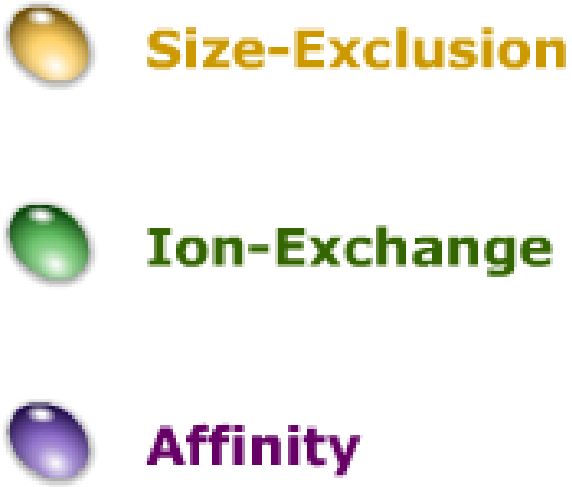
a) net charge
b) solubility in salt solutions
c) size or mass

17) In ion-exchange chromatography

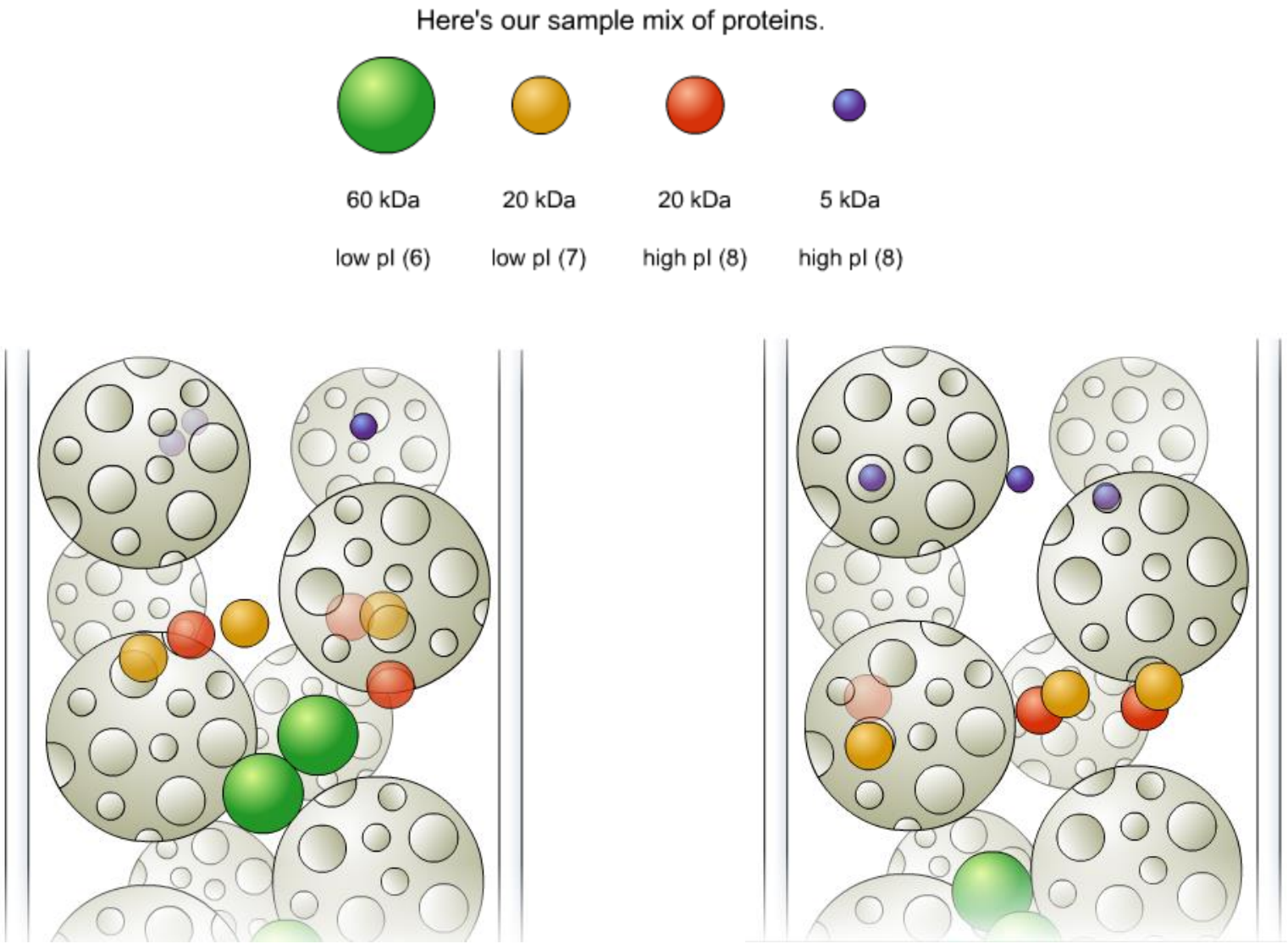
a) proteins are separated on the basis of their net charge

Most common method for separating proteins

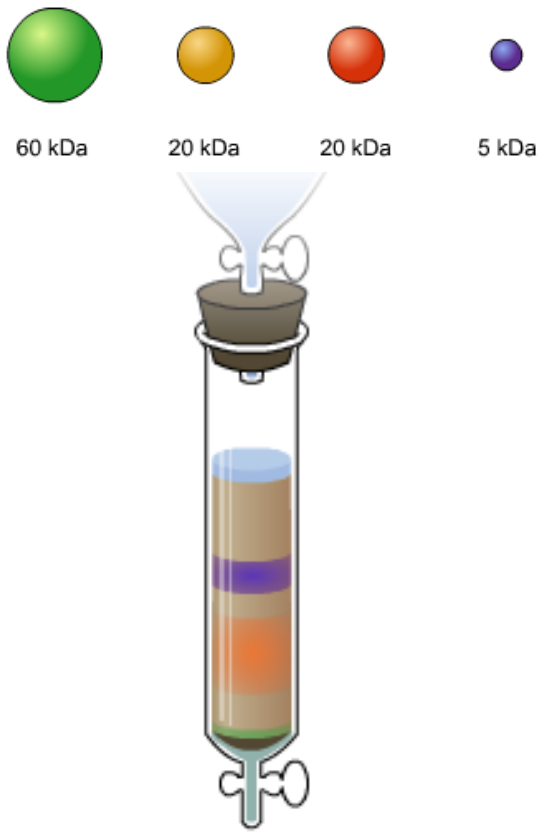
Column Chromatography



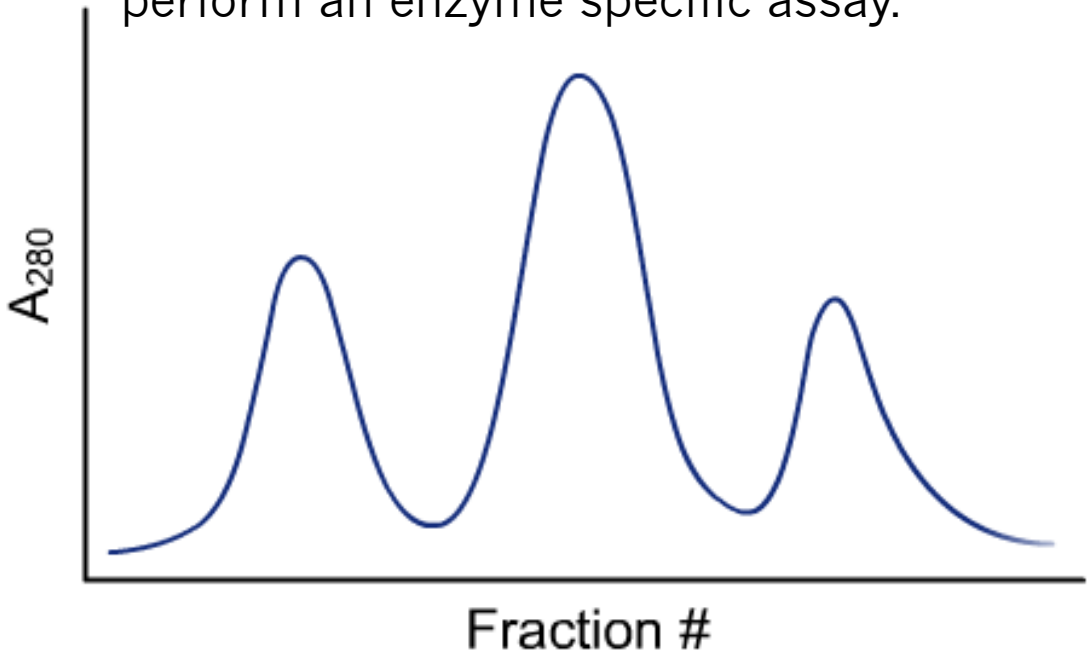
Size-exclusion chromatography

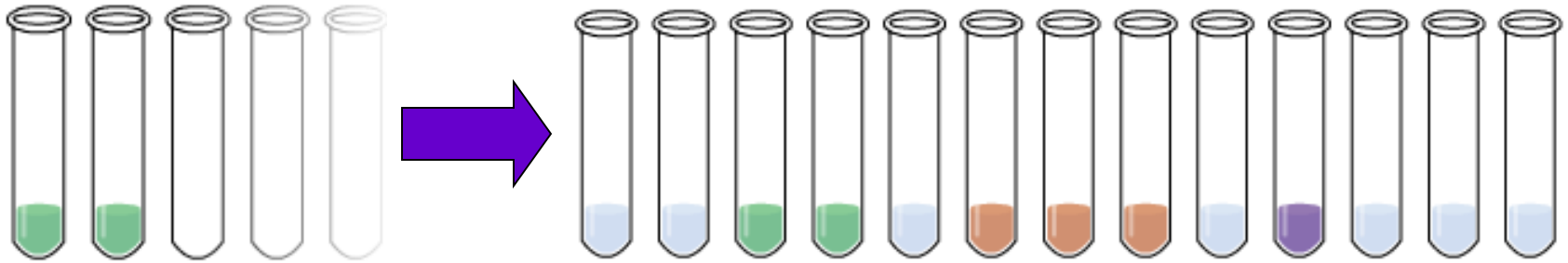


Size-exclusion chromatography

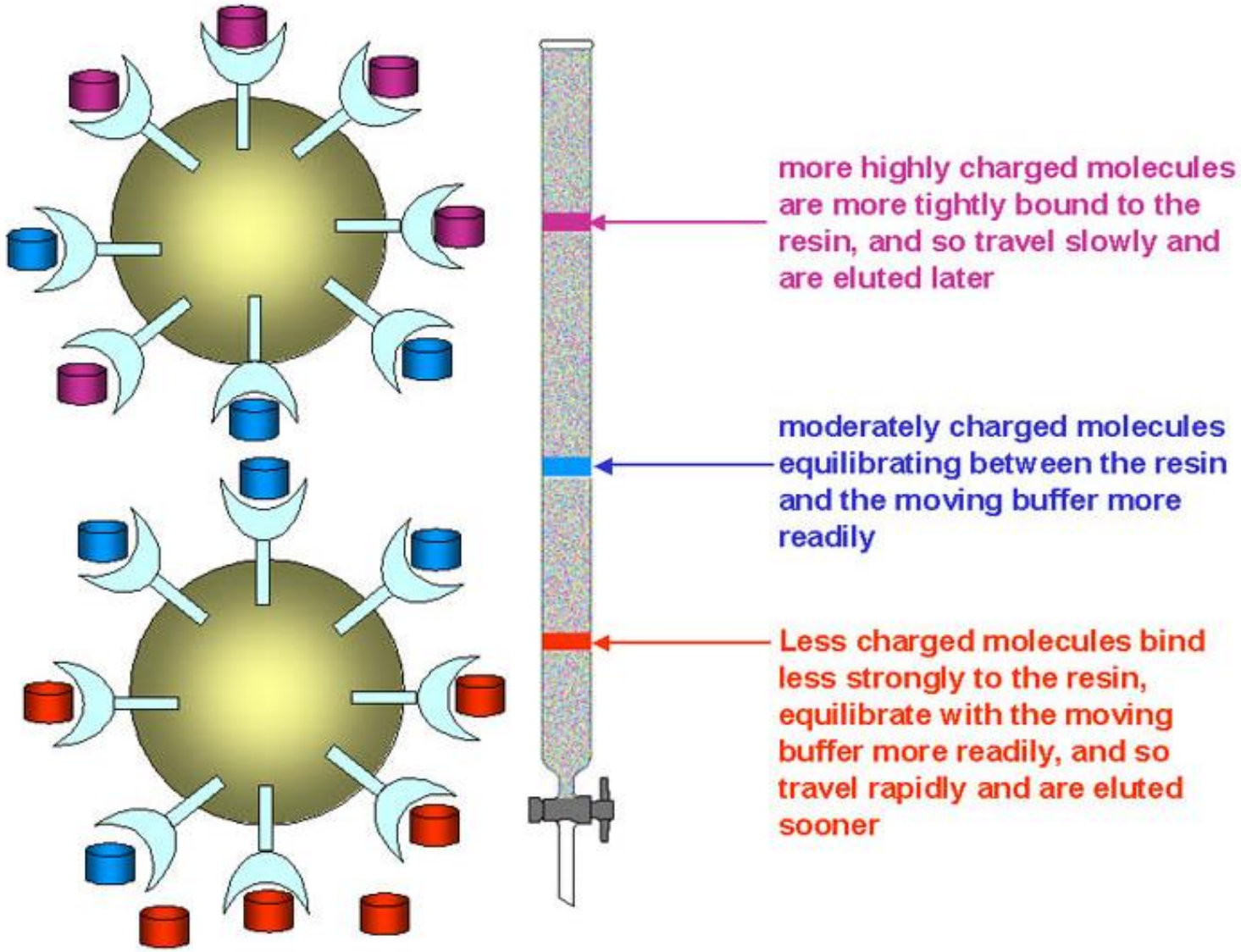


Absorbance at 280 is used to identify protein-containing fractions. You can also perform an enzyme specific assay.





ANION EXCHANGE CHROMATOGRAPHY



Cation Anion Exchange Chromatography

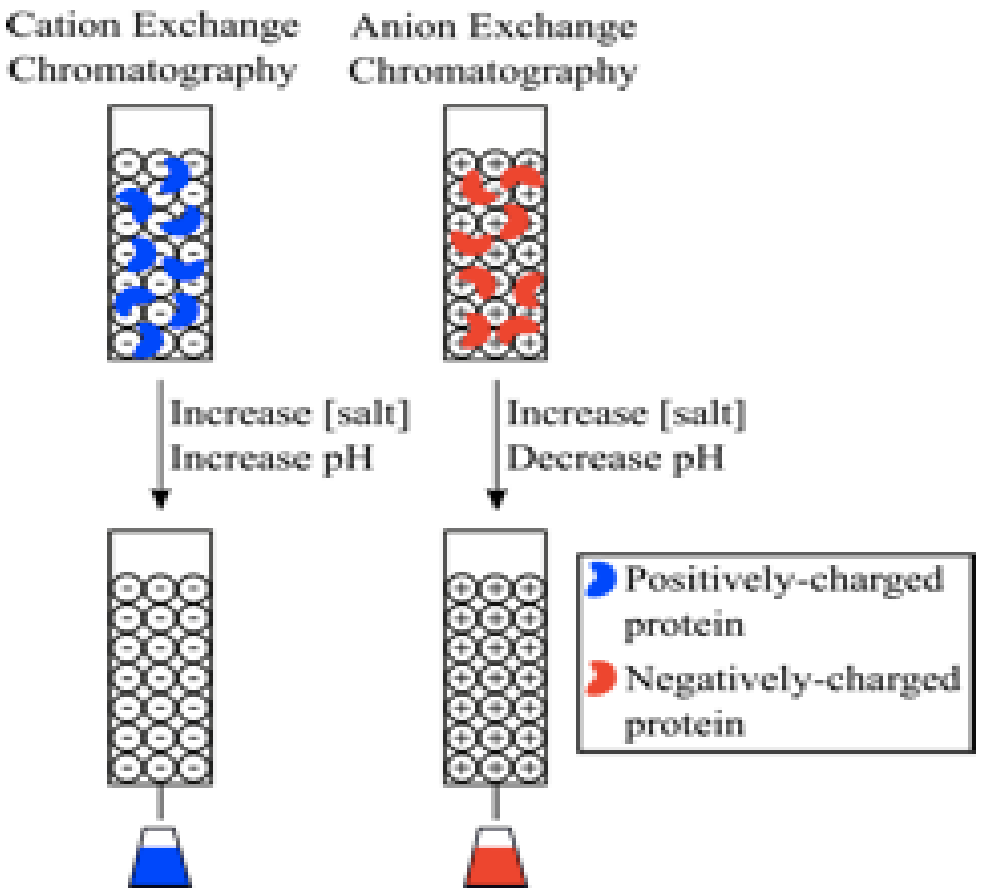


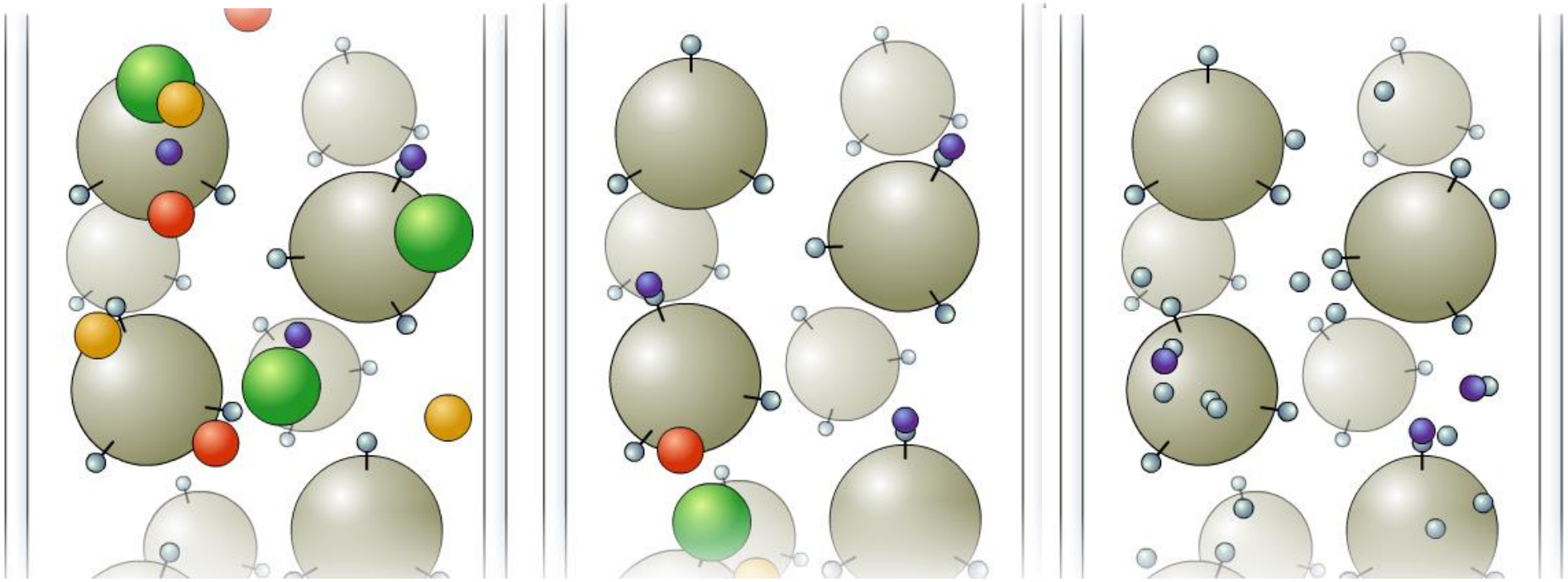
Figure 6. Ion exchange chromatography. Proteins are bound to a charged stationary phase at low ionic strength. The bound proteins can be eluted either by increasing the ionic strength of the buffer or by adjusting the pH.

Affinity chromatography

Makes use of specific binding interactions between molecules

1- Incubate crude sample with the immobilized ligand

3- Elute



2- Wash away non bound sample components from solid support